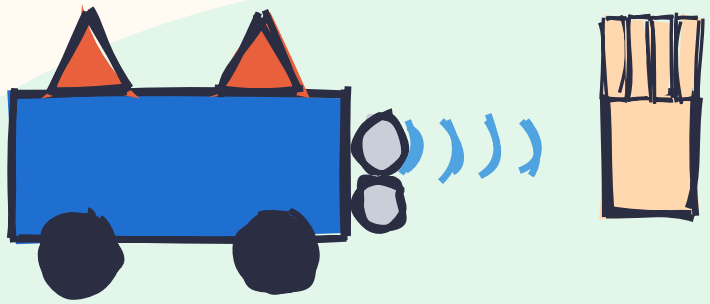
Little Explorers · Class 30

Follow-Me Robot Pet

A robot that follows your hand like a puppy!

Eyes | Rules | Pet



Did you know?
Some robot dogs can follow their owner and even play fetch!

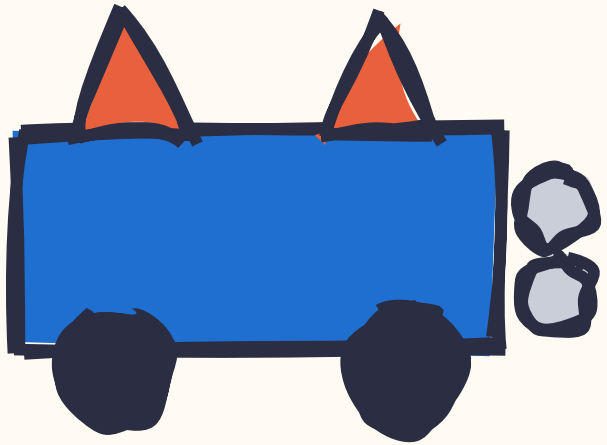
LEARN

Let's wonder...

How does a puppy follow you?

Here, boy!

Walk away and your puppy trots after you. Stop, and it stops. It **sees** you, then **decides** what to do. Can a robot car do the same?



It can't!

It needs a leash?

It needs eyes!

Today: you turn a robot car into a **pet** that follows your hand and stops before it bumps you.

LEARN

Learning objectives

Today I can…

1		**Robot eyes:** say that the robot's **eyes** are a sensor that measures how far away my hand is	**Proof:** I showed where the robot's eyes are
2		**Near and far rules:** use rules: **far = come closer**, **too near = stop**	**Proof:** my robot stopped before it bumped my hand
3		**Pet personality:** give my robot pet a **name and a costume** and show it to the class	**Proof:** my pet followed me in the parade

Objective 1 · Robot eyes · video

Video time!



Sony's robot dog Aibo vs. a real puppy

Washington Post · about 4 minutes · youtu.be/tLjtOMBWvTo

Watch for...

1. How is the robot dog like a real puppy?
2. What can a real puppy do that the robot can't?
3. How does the robot know where you are?

Scan to watch

After the video: Would you rather have a robot pet or a real pet? Why?

Objective 1 · Robot eyes

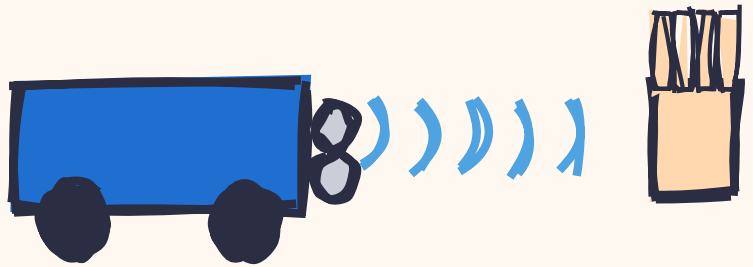
The robot's eyes measure distance

The idea

The **ultrasonic sensor** on the front looks like two eyes. One eye sends a tiny sound **'ping'** (too high for us to hear). The other eye listens for the **echo**.

Quick echo = **near**. Slow echo = **far**.

Example: a bat in the dark



Bats squeak and listen for echoes to find bugs and walls in the dark. Our robot pet 'sees' your hand the same way!

LEARN

Objective 2 · Near and far rules

Near and far rules

The idea

The robot pet follows **rules**:
Far away → drive forward (come here!)
Just right → stop and wait
Too near → back up a little

The rules repeat **forever**, again and again, very fast.

Example: playing tag

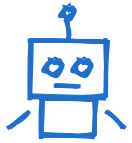


When you chase a friend you keep checking: far? run! close? slow down so you don't crash! Robots check like that too.

LEARN

Real world

Robots that follow



Robot dogs
Follow their owner and play



Shopping carts
Some carry bags and follow you



Factory robots
Follow lines and people safely



Smart cars
Keep a safe gap from the car in front

? What job could a follow-me robot do at school?

Safe gap
Smart cars use the same idea: if the car ahead is too near, slow down!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Puppy and owner
2. What we use
3. Snap the picture blocks
4. Wake up your pet
5. Change the rules
6. Dress up your pet
7. Pet tests

Class game

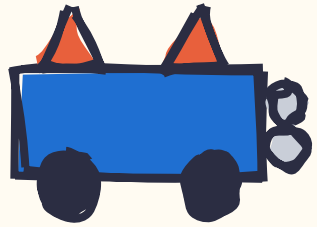
Puppy and owner

How to play

1. Pairs: one is the **owner**, one is the **puppy**.
2. Owner holds a hand up and walks slowly.
3. Puppy rules: **far = walk**, **one big step away = stop**, **too close = step back**.
4. Swap! Then try with 3 puppies in a line.

Then discuss

What rules did the puppy use to keep a safe gap?



Big idea

The robot pet uses the same three rules, forever.

Build it

What we use

Robot car
with ultrasonic eyes

mBlock 5
starter program, teacher-uploaded

Costume
paper ears, tail, face

Tape & pens
to decorate

Teacher uploads the starter program before class.

Gentle!
The robot is a pet: pick it up with two hands and never drop it.

MAKE

Time to build!

Snap the picture blocks

```
forever
    if distance > 30 then
        move forward
    else if distance < 10 then
        move backward
    else
        stop
```

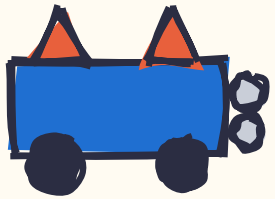
Read it like a story

Look with the eyes.

Far? Come closer.

Too near? Back up.

Otherwise: **stop** and wait.



Number = distance

30 means about 30 cm: the length of a ruler!

Objective 2 · Build step 1 of 3

Wake up your pet

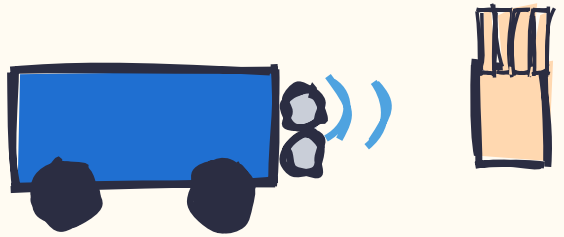
a Switch on the robot car. Teacher has loaded the program.

b Put it on the floor with lots of space.

c Hold your hand in front of its eyes, far away.

d Does it drive towards you?

e Move your hand closer. Does it stop?



Check: Your pet comes when far and stops when near.

Keep fingers away from the wheels.

MAKE

Objective 2 · Build step 2 of 3

Change the rules

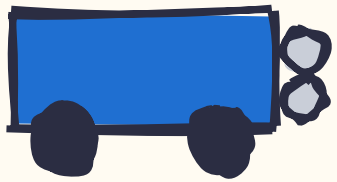
a With the teacher, change **30** to **50**. What happens?

b Change it to **15**. What happens now?

c Add a **sound**: play a beep when it stops (like a bark!).

d Add a **light**: LED on when it sees you.

e Test after every change.



beep!

Check: Your pet barks when it reaches you.

Objective 3 · Build step 3 of 3

Dress up your pet

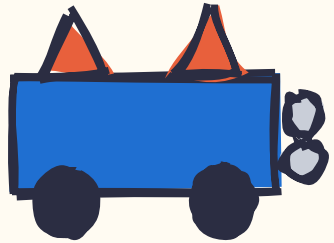
a Choose an animal: puppy, kitten, dinosaur, bunny…

b Make ears, a tail and a face from paper.

c Tape them on. **Don't cover the eyes** (the sensor)!

d Give your pet a **name**.

e Get ready for the pet parade!



Check: Your pet has a costume and a name.

Test it!

Pet tests

Test	I think it will…	What happened
Hand far away		
Hand very close		
A book instead of a hand		

Talk about it

Did your pet follow a book too? Why?

Eyes free
If the costume covers the sensor, the pet goes blind!

Level up!

Pet trainer challenges

Choose your challenge



Pet parade: lead your pet around a cone and back.



Train of pets: can one pet follow another pet?



Happy tune: play a song when it reaches you.



Shy pet: make a pet that runs AWAY when you come near!

Think like an engineer

Is a robot pet as good as a real pet? What can't it do?

Boss level

Teach your pet a trick: spin when you're very near!

Objective check

Can I really do it? Prove it.

01	Point to your robot's eyes. How do they work?	○ ○ ○
02	Tell me the 3 rules your pet follows.	○ ○ ○
03	Show your pet following you in the parade.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Pet parade: each pet follows its owner across the room!

Show what you know

What did we learn?

- [x] The ultrasonic eyes measure distance with echoes
- [x] Far = come closer, too near = back up
- [x] Rules repeat forever, very fast
- [x] Changing a number changes the behaviour

Exit ticket: what does the robot pet do when your hand is far away?

At home

Watch a pet or a little brother follow someone. What 'rules' do they use?



Next class: Safety Helper Robot!